

① Possible splits = 3, 5, 7, 9

Pass = 1
fail = 0

① Split = 3

→ $G_1 = (S.H. < 3)$; Students: 2,
fail = 1 Pass = 0.

$$\text{Gini}(G_1) = 1 - (1^2 + 0) = 0$$

→ $G_2 = (S.H. > 3)$; Students: 4, 6, 8, 10.

fail = 1 Pass = 3

$$\text{fail} = \frac{1}{4} \quad \text{Pass} = \frac{3}{4}$$

$$\begin{aligned} \text{Gini}(G_2) &= 1 - \left(\frac{1^2}{4} + \frac{3^2}{4} \right) = 1 - \left(\frac{1}{16} + \frac{9}{16} \right) \\ &= 0.375 \end{aligned}$$

$$\rightarrow \text{Weighted Gini} = \frac{1}{5} \cdot 0 + \frac{4}{5} \cdot 0.375 = 0.3$$

② Split = 5

→ $G_1 = (S.H. < 5)$; Students: 2, 4.

fail = 2, Pass = 0

$$\text{Gini}(G_1) = 1 - \left(\frac{2^2}{2} + 0^2 \right) = 0.$$

→ $G_2 = (S.H. > 5)$; Students: 6, 8, 10

fail = 0, Pass = 3.

$$\text{Gini}(G_2) = 1 - (0^2 + \frac{3^2}{3}) = 0.$$

$$\rightarrow \text{w. G} = \frac{2}{5} \cdot 0 + \frac{3}{5} \cdot 0 = 0.$$

③ Split = 7

→ $G_1 = (S.H. < 7)$; Students: 2, 4, 6.

classmate
Date _____
Page _____

$$\text{fail} = 2, \quad \text{Pass} = 1$$

$$\text{Gini}(G_1) = 1 - \left(\frac{2^2}{3} + \frac{1^2}{3} \right) = 0.444$$

$$\rightarrow G_2 = (SH > 7); \text{ Students} = 8, 10$$

$$\text{Pass} = 2, \quad \text{fail} = 0$$

$$\text{Gini}(G_2) = 1 - \left(\frac{2^2}{2} + 0^2 \right) = 0$$

$$\rightarrow \text{weighted Gini} = \frac{3}{5} \cdot 0.444 + \frac{2}{5} \cdot 0 = 0.266$$

⑩ Split = 9

$$\rightarrow G_1 = (SH < 9); \text{ Students} = 2, 4, 6, 8$$

$$\text{Pass} = 2, \quad \text{fail} = 2$$

$$\text{Gini}(G_1) = 1 - \left(\frac{2^2}{2} + \frac{2^2}{2} \right) = 0.5$$

$$\rightarrow G_2 = (SH > 10); \text{ Students} = 10$$

$$\text{Pass} = 1, \quad \text{fail} = 0$$

$$\text{Gini}(G_2) = 0$$

$$\rightarrow \text{weighted Gini} = \frac{4}{5} \cdot 0.5 + \frac{1}{5} \cdot 0 = 0.4$$

$\therefore$ In split 5 weighted Gini is 0
This is lowest weighted Gini if we compared to other splits.

$\therefore$ Prediction is

if Study hours $\leq 5 \rightarrow$ Prediction Fail (0)
if Study hours $> 5 \rightarrow$ Prediction Pass (1)